lesson 11 Counting
(Combinatorial Proofs)

(Ordered) list of k elements
(length k)

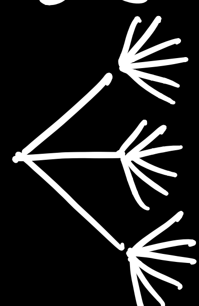
$(a_1, a_2, \ldots, a_k)$

- list is ordered : $(1,2) \neq (2,1)$
  set is unordered : $\{1,2\} = \{2,1\}$
- repetitions possible in list :
  $(1,1) \neq (1)$
  set is non-repetitions :
  $\{1,1\} = \{1\}$

k-element list : $(a_1, a_2, \ldots, a_k)$
each element a_j has n_j possibilities.

$\#$ of lists $= n_1 n_2 \cdots n_k$
Multiplication Principle (MP)

(pants, shirt) $3 \cdot 6 = 18$
3 6



Prop 1 A, B : sets, $|A| = n$, $|B| = m$.
$|A \times B| = nm$.

Proof $A \times B$ is the set of 2-element lists (x,y) s.t. $x \in A$ and $y \in B$. There are n possibilities for x and m possibilities for y. By MP, there are nm elements (lists) in $A \times B$.

k-element list $(a_1, \ldots, a_k)$

each element is selected from the <u>same</u> n objects.

(i) repetitions are allowed:

$$n_1 = n_2 = \cdots = n_k = n$$

$$\# \text{lists} = \underbrace{n \cdot n \cdot \cdots n}_{k} = n^k$$

(ii) <u>no</u> repetitions:

$$\boxed{(n-k+1)}$$

$$\# \text{lists} = \underbrace{n(n-1)(n-2)\cdots(n-(k-1))}_{k}$$

$$\begin{array}{ccccc} n-0 & n-1 & n-2 & \cdots & n-(k-1) \\ 0 & 1 & 2 & \cdots & k-1 \end{array}$$

$$\underbrace{0 \quad 1 \quad 2 \quad \cdots \quad k-1}_{k}$$

$$\boxed{P(n,k) = n(n-1)(n-2)\cdots(n-k+1)}$$

<u>permutation</u>
(arrangement)

$$P(n,k) = \#\text{ of non-repetitions } k\text{-element lists}$$

where each element has the same n possibilities

$$= \#\text{ of ways to } \underline{\text{arrange}}\ \underline{k}\ \underline{\text{of}}\ \underline{n} \text{ objects}$$

$$\overbrace{\underline{\ \ }\ \ \underline{\ \ }\ \ \underline{\ \ }\ \cdots\ \underline{\ \ }}^{k}$$
$$n \quad n-1 \quad n-2 \qquad\qquad n-k+1$$

<u>Ex</u> password, 5 lower case letters

a) # passwords:
$26 \times 26 \times 26 \times 26 \times 26 = 26^5$

b) distinct letters:
$26 \times 25 \times 24 \times 23 \times 22 = P(26,5)$

c) ends in a <u>vowel</u> (a, e, i, o, u):
$26 \times 26 \times 26 \times 26 \times 5 = 26^4 \cdot 5$

d) w/out any vowel
$21^5 = 21 \times 21 \cdot 21 \cdot 21 \cdot 21$

e) <u>at least one vowel</u>

(complement of no vowels)

$$\underbrace{26^5}_{\text{all}} - \underbrace{21^5}_{\text{no vowels}}$$

$n \in \mathbb{N}$, <u>n factorial</u>

$$n! = n(n-1)(n-2)\cdots 3\cdot 2\cdot 1$$

$$0! = 1$$

$$P(n,k) = \frac{n(n-1)(n-2)\cdots(n-k+1)\cancel{(n-k)\cdots 3\cdot 2\cdot 1}}{\cancel{(n-k)(n-k-1)\cdots 3\cdot 2\cdot 1}}$$

$$P(n,k) = \frac{n!}{(n-k)!}$$

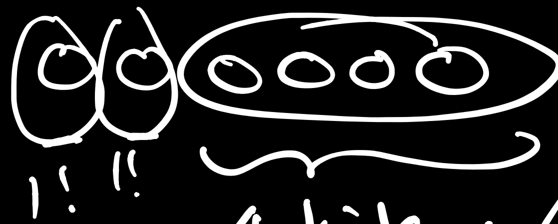
$$P(n,n) = \frac{n!}{(n-n)!} = \frac{n!}{\underbrace{0!}_{1}} = n!$$

$n! =$ # of ways to arrange n objects (in a line)

<u>Ex</u> 2 parents, 4 kids

kids sit together

how many ways to arrange?



3 groups : $3!$

4 kids : $4!$

Ans : $3!\,4!$

^nP_k

$P(n,k)$ = # of ways to arrange k of n objects — order matters

Combination

$C(n,k)$ = ① # of ways to choose k of n objects

$\binom{n}{k}$ n choose k — order does not matter

^nC_k

= ② # of k-element subsets of an n-element set

$$\boxed{\underbrace{P(n,k)}_{\text{arranging k of n objects}} = \underbrace{\binom{n}{k}}_{\text{choosing k of n objects}} \underbrace{k!}_{\text{arranging k objects}}}$$

$$\binom{n}{k} = \frac{P(n,k)}{k!}$$

$$\boxed{\binom{n}{k} = \frac{n!}{(n-k)!\,k!}}$$

Ex password : 6 digits
0, 1, ..., 9

How many passwords have exactly two 5's ?

$\_\ \_\ \underline{x}\ \_\ \underline{x}\ \_$

first, choose 2 of 6 places to put "5" : $\binom{6}{2}$

next, put a non-5 in every of the 4 remaining places.

$9 \times 9 \times 9 \times 9 = 9^4$

Ans: $\binom{6}{2} 9^4$